

Referee-Style Feedback

Manuscript

Galdino, Manoel, Davi Moreira, and Carolina Dolleães. “Included Variable Bias in Dynamic Panels: A Diagnostic for Time-Varying Controls and a Benchmark for the Contemporaneous Treatment Effect.” July 2026.

Overall Assessment

This is a **promising and potentially useful methods paper**, especially for applied political scientists working with time-series cross-sectional data. Its central intuition is important, its practical recommendation is memorable, and the manuscript is unusually clear about the limits of that recommendation.

My referee-style recommendation would be **major revision**. The paper has a publishable core, but it currently moves too quickly from an algebraic identity to the language of “bias,” and from a set of simulation designs to a fairly general ADL plus fixed-effects benchmark. The revision should more sharply separate three analytical tasks:

1. **Causal classification:** What role does the covariate play in the causal DAG?
2. **Descriptive diagnosis:** How much does adding the covariate move the treatment coefficient?
3. **Identification:** Under what assumptions does either coefficient recover the contemporaneous causal effect?

This distinction is already implicit in the manuscript, but it should become the paper’s formal organizing principle.

1. Summary of the Paper

The manuscript studies the problem of conditioning on time-varying covariates that may themselves respond to treatment. Its target setting is a panel or time-series cross-sectional design in which the researcher wants the contemporaneous effect of D_t on Y_t , but must decide whether to control for Z_t or Z_{t-1} .

The paper makes two connected claims.

First, it introduces **included variable bias**, or IVB, as a diagnostic for the coefficient shift produced by adding a candidate control. For nested linear models, the manuscript uses the Frisch–Waugh–Lovell identity

$$\hat{\beta}^* - \hat{\beta} = -\hat{\theta}^* \hat{\pi},$$

where:

- $\hat{\theta}^*$ is the coefficient on the candidate control in the long outcome regression;
- $\hat{\pi}$ is the coefficient on treatment in an auxiliary regression for that control.

The authors correctly emphasize that the formula does not, by itself, establish whether the coefficient change is harmful. The causal DAG determines whether the change represents collider bias, over-control, or deconfounding.

Second, the paper recommends **autoregressive distributed-lag models with unit and time fixed effects and lagged state variables**—ADL plus FE—as a practical starting point when the estimand is the contemporaneous treatment effect. Contemporaneous controls are treated as suspect when they may be mediators or colliders, while substantively justified lagged controls can enter as predetermined state variables.

The simulations cover three principal situations:

- a contemporaneous collider;
- a dual-role covariate that is useful when lagged but problematic contemporaneously;
- a contemporaneous mediator, sometimes combined with lagged confounding.

The applications calculate IVB quantities for controls used in published political-science studies, including GDP per capita in research on democracy and ethnic inequality and in research on postal infrastructure and economic development.

2. Contribution to Political Methodology

2.1 Why the Literature Should Care

The practical problem addressed by the paper is real and consequential. Political scientists regularly include variables such as GDP, repression, conflict, turnout, aid, inflation, or institutional quality without making the within-period causal ordering explicit.

A variable can be a legitimate pretreatment confounder at one time index and a mediator or collider at another. The manuscript gives researchers a simple language for discussing this issue and a workflow that can be implemented using familiar regression models.

The paper is especially valuable as a bridge between three conversations that applied researchers often encounter separately:

1. causal-DAG discussions of good and bad controls;
2. panel-data discussions of state dependence and dynamic specifications;
3. recent work on treatment-responsive, time-varying covariates.

The manuscript's strongest conceptual statement can be summarized as follows:

DAGs determine the causal meaning of a specification change; IVB measures the size of that change; and the dynamic benchmark provides an applied comparison.

This is clear, teachable, and potentially influential.

2.2 What Is Genuinely Distinctive

The strongest contribution is **not the algebraic identity by itself**. The decomposition is a standard nested-regression result. A skeptical reviewer may therefore argue that “included variable bias” merely renames the difference between short- and long-regression coefficients.

The more defensible contribution is the complete diagnostic framework:

1. use timing and a dynamic DAG to classify the candidate control;
2. decompose the coefficient shift into the treatment–control and control–outcome relationships;
3. compare contemporaneous and lagged versions of the control inside an explicitly dynamic specification;
4. connect the recommendation to a clearly defined estimand—the contemporaneous treatment effect.

The paper will be stronger if it claims this **diagnostic workflow** as its principal innovation, rather than implying that the product formula itself is a major new statistical result.

2.3 Placement in the Literature

Relative to the literature cited in the manuscript, the paper occupies a useful middle position.

It is less general than work on sequential identification, time-varying confounding, or treatment-responsive covariates, but it is more operational for conventional linear TSCS research. It does not solve the problem of recovering an untreated potential covariate, and it does not replace causal identification assumptions.

Instead, it offers:

- a specification-sensitivity diagnostic;
- a way to interpret that diagnostic using causal graphs;
- a baseline specification for researchers who remain within linear fixed-effects models.

That is a legitimate contribution. The paper should, however, frame itself primarily as an **applied diagnostic and specification-auditing framework**, rather than as a general solution to time-varying confounding.

3. The Key Figure

Figure 2 as the Empirical Centerpiece

Figure 2 is the manuscript’s most important empirical representation.

It presents the dual-role simulation in which ADL plus fixed effects with Z_{t-1} maintains bias around 0.008 to 0.010 across values of covariate persistence. By contrast, TWFE with Z_{t-1} remains substantially biased, and the static short specification performs worst.

The associated analysis argues that outcome persistence transmits an inherited collider association from Y_{t-1} to Y_t , while conditioning on the lagged outcome blocks that transmission.

The figure therefore does more than demonstrate that “lagging the control helps.” It supports the more theoretically interesting proposition that:

A lagged bad-control path may continue to affect the contemporaneous treatment equation when the outcome is persistent, so the lagged outcome is performing substantive causal work in the proposed benchmark.

Table 2 reinforces this interpretation. When outcome persistence is zero, TWFE with Z_{t-1} produces almost no bias. When persistence is positive, TWFE bias becomes large, while the ADL specification remains close to zero.

Recommended Improvements to Figure 2

The figure should include:

- a more prominent zero-bias reference line;
- Monte Carlo uncertainty intervals;
- the number of simulations;
- the panel dimensions N and T ;
- either relative bias or RMSE in addition to raw bias;
- clearer visual differentiation among the specifications.

Most importantly, the manuscript should place the relevant **dynamic DAG** immediately before the figure. The generic collider–fork–chain diagram does not show the inherited path that Figure 2 is intended to illuminate.

4. Why the Research Is Challenging and Novel

The research problem is difficult because the causal role of Z cannot be determined from the substantive name of the variable alone.

“GDP,” “conflict,” “repression,” or “institutional quality” is not inherently a confounder, mediator, or collider. Its role depends on:

- the treatment and outcome being studied;
- the time index;
- the within-period causal ordering;
- persistence and feedback across periods;
- the causal estimand.

The same covariate may therefore be useful at $t - 1$ and harmful at t . It may also play more than one role. Lagged GDP, for example, can capture baseline developmental conditions, while contemporaneous or future GDP may partly result from treatment and accumulated outcomes.

The manuscript is novel insofar as it seeks to make this ambiguity operational rather than simply warning researchers about “bad controls.”

The simulations also highlight a less obvious point: lagging a collider does not necessarily eliminate its consequences when the outcome process carries the induced association forward.

5. Major Limitations and Revision Priorities

5.1 A Specification Shift Is Not Automatically Bias

This is the most important conceptual issue.

The identity

$$\hat{\beta}^* - \hat{\beta} = -\hat{\theta}^* \hat{\pi}$$

shows how much two nested regression coefficients differ. It does not establish:

- which coefficient is biased;
- whether both coefficients are biased;
- whether either coefficient identifies a causal effect;
- whether the two coefficients target the same estimand.

The manuscript acknowledges this issue verbally, but the terminology continues to invite a causal interpretation.

For example:

- when Z is a confounder, the shift may represent bias reduction;
- when Z is a mediator, the regressions may target total and direct effects;
- when Z is a collider, the long coefficient may be biased;
- under unobserved confounding or model misspecification, neither coefficient may identify the contemporaneous treatment effect.

Recommendation

Formally define IVB first as a **population and sample specification-shift diagnostic**. Reserve the word “bias” for settings in which a causal DAG and explicit identification assumptions establish a target parameter and demonstrate that conditioning on Z causes deviation from that parameter.

The notation should distinguish:

$$\Delta_Z = \beta_{\text{long}} - \beta_{\text{short}}$$

from:

$$\text{Bias}(\hat{\beta}) = E[\hat{\beta}] - \beta_{\text{CET}}.$$

The first quantity is defined for nested linear projections. The second requires a causal estimand and identification assumptions.

One possible solution is to use **included-variable specification shift** as the general term and reserve **included variable bias** for causally classified cases.

5.2 The ADL Benchmark Needs Explicit Identification Conditions

The manuscript presents ADL plus fixed effects as a “defensible baseline.” That is reasonable, but readers need a precise statement of the assumptions under which its contemporaneous treatment coefficient has a causal interpretation.

At minimum, the paper should discuss:

- sequential or conditional exogeneity of D_t ;
- whether treatment can respond to past outcomes;
- the role of D_{t-1} and longer treatment histories;
- restrictions on contemporaneous feedback between D_t and Y_t ;
- correct specification of the lag structure;
- unit-specific trends;
- heterogeneous dynamics;
- the required relationship between N and T ;
- serial correlation in the error process;
- appropriate inference under cross-sectional dependence.

The paper’s own simulations show that the benchmark does not solve contemporaneous unobserved confounding. In those designs, bias reaches between 8 and 53 percent of the treatment effect.

This limitation should appear earlier and more prominently. Ideally, it should be incorporated into the formal statement introducing the benchmark.

The benchmark should be framed as:

a specification baseline that addresses particular included-variable and timing problems under stated sequential-identification assumptions,

rather than as a generally causal panel estimator.

5.3 Dynamic Panel Bias Is Largely Missing

An ADL model with fixed effects and a lagged dependent variable can suffer from finite- T dynamic panel bias. Whether this problem is negligible depends on the time dimension, persistence, and estimator.

This matters because the paper is explicitly aimed at TSCS researchers, whose panels sometimes have short or moderately sized time dimensions.

Recommendation

Add simulations varying:

- N ;
- T ;
- outcome persistence ρ_Y ;
- treatment persistence;
- the magnitude of unit effects.

The simulations should compare, where appropriate:

- the conventional fixed-effects within estimator;
- bias-corrected fixed effects;
- a suitable GMM estimator;
- alternative dynamic estimators.

The paper does not need to endorse one estimator universally. It should, however, distinguish two questions:

1. Which variables should enter the conditioning set?
2. How should a dynamic model containing Y_{t-1} be estimated?

The current manuscript addresses the first question but risks obscuring the second.

5.4 “Contemporaneous” Timing Requires a Stronger Substantive Definition

In annual political data, D_t , Z_t , and Y_t are often aggregates over the same interval. Calling Z_t post-treatment or Z_{t-1} pretreatment can therefore be too mechanical.

For example:

- if treatment occurs near the end of year t , much of Z_t may precede treatment;
- if treatment is anticipated, Z_{t-1} may already respond to future treatment;
- if treatment is continuous during the year, the causal sequence may include repeated feedback;
- if the outcome is measured at the beginning rather than the end of the period, the assumed ordering may reverse.

Recommendation

Define a within-period temporal model explicitly, for example:

$$\text{state at beginning of } t \rightarrow D_t \rightarrow Z_t \rightarrow Y_t.$$

Then explain how annual, quarterly, or otherwise aggregated observations map onto these causal moments.

The benchmark should be conditional on this substantive temporal interpretation, not merely on variable subscripts.

5.5 The Simulations Are Insufficiently Documented

The reported simulation results are striking, but the manuscript does not provide enough information in the main text to assess or reproduce them.

Readers need:

- complete data-generating equations;
- all parameter values;
- values of N and T ;
- burn-in procedures;
- number of replications;
- treatment distributions;
- error distributions;
- fixed-effect generation;
- initial conditions;
- exact estimator implementation;
- bias definitions;
- variance and RMSE measures;
- inferential procedures;
- replication code.

Without these details, the simulations may appear designed to validate the preferred benchmark.

Recommendation

Provide a complete simulation appendix and a summary table connecting each simulation design to:

- the corresponding DAG;
- the causal estimand;
- the identifying assumptions;
- the estimators being compared;
- the expected source of bias.

The paper should also vary the strength of each causal arrow independently, rather than concentrating mainly on persistence parameters. This would reveal the parameter regions in which the proposed benchmark performs well or fails.

5.6 The DAG Presentation Is Too Generic

Figure 1 presents elementary collider, fork, and chain structures. The paper's substantive results, however, concern dynamic DAGs involving:

- temporal feedback;
- persistence;
- inherited collider paths;
- lagged conditioning;
- multiple causal roles for the same substantive variable.

The generic diagrams are not sufficient to establish the more complex claims.

Recommendation

Add at least three fully time-indexed DAGs:

1. a contemporaneous collider design;
2. a dual-role design in which Z_{t-1} is a confounder and Z_t is a collider;
3. a design in which Z_t is a mediator and Z_{t-1} is a lagged confounder.

For the dual-role design, explicitly trace the inherited path and show why conditioning on Y_{t-1} blocks its transmission to Y_t .

This mechanism is one of the paper's most interesting theoretical insights. Readers should not have to reconstruct it independently.

5.7 Multiple Controls Require a Multivariate Treatment

The scalar decomposition is clear, but applied regressions normally contain several correlated controls.

For a vector of candidate controls $\mathbf{Z}$, the corresponding decomposition is:

$$\Delta\beta = -\theta'\pi.$$

One-at-a-time IVB quantities can be misleading when controls are correlated. A variable may appear unimportant because another control has already absorbed the relevant variation. The value assigned to a control may also depend on which other controls are included.

Recommendation

Add:

- the vector decomposition;
- a joint IVB measure;
- leave-one-control-out diagnostics;
- discussion of order dependence;
- diagnostics for correlated controls.

A Shapley-style decomposition could be considered if the authors want to allocate a joint specification shift among multiple controls. This is not essential, but the manuscript must at least explain why one-at-a-time quantities may not sum to the total shift.

This issue is directly relevant to the empirical applications, which consider several controls drawn from the same original specification.

5.8 Statistical Uncertainty for IVB Needs Development

The use of $|IVB/SE|$ is intuitive but underdefined.

The paper should clarify:

- which standard error appears in the denominator;
- whether it is the short-model treatment SE;
- whether it is the long-model treatment SE;
- whether it is the SE of the difference between coefficients;
- how covariance between the nested estimators is handled;
- whether uncertainty in $\hat{\theta}$ and $\hat{\pi}$ is propagated.

Because the sample decomposition is algebraically exact, the relevant uncertainty concerns the underlying population specification shift.

Recommendation

Develop a principled uncertainty measure using:

- a delta-method variance;
- a joint sandwich estimator;
- or a cluster bootstrap appropriate to the panel structure.

A diagnostic table could report:

$$\widehat{\Delta}_Z, \quad CI(\Delta_Z), \quad \frac{|\widehat{\Delta}_Z|}{|\hat{\beta}_{\text{benchmark}}|},$$

along with a substantively meaningful threshold.

The paper should avoid relying exclusively on a comparison with a single regression standard error.

5.9 The Applications Need a Clearer Sampling and Coding Protocol

The manuscript reports 14 candidate controls across six studies, with a median $|IVB/SE|$ of 0.13 and only one case above one standard error. However, only two applications receive detailed treatment.

The authors should explain:

- how the six studies were selected;
- whether they represent a systematic sample;
- how candidate colliders were identified;
- who performed the causal classification;
- whether classifications were made before examining coefficient changes;
- whether multiple coders were used;
- whether original samples were reproduced exactly;
- whether original standard errors and clustering procedures were replicated;
- why each variable was classified as a collider, mediator, confounder, or dual-role variable;
- what happened in all 14 cases.

The relatively small median shift is not a weakness. It changes the practical interpretation of the contribution.

The evidence suggests that IVB may be most valuable as a **screening, transparency, and specification-auditing device**, rather than because bad controls routinely reverse published findings.

That is still an important contribution, but the paper should state it directly.

5.10 The Substantive DAGs in the Applications Need Tightening

The GDP applications are plausible, but the causal arguments remain somewhat verbal.

In the postal-infrastructure application, for example, the reader needs a time-indexed explanation of how lagged GDP becomes a collider rather than merely a confounder affected by previous treatment.

The Rogowski et al. application produces a large specification shift—approximately 58 percent of the long coefficient and 2.11 standard errors—but the causal meaning remains ambiguous.

That ambiguity is not a failure of the diagnostic. It is an effective illustration of its proper use.

Recommendation

For each detailed application, present at least two competing causal DAGs:

1. a DAG in which the candidate control is a legitimate confounder;
2. a DAG in which it is treatment-responsive, a mediator, a collider, or a dual-role variable.

Then show that the same arithmetic IVB value receives different causal interpretations under the competing structures.

This would demonstrate that IVB is a diagnostic requiring substantive causal reasoning, rather than an automatic variable-selection criterion.

6. Contexts in Which the Paper Is Less Applicable

The proposed benchmark is least reliable when:

- the estimand is cumulative, delayed, distributed, or long-run rather than contemporaneous;
- treatment timing within the period is unclear;
- contemporaneous unobserved confounding is important;
- anticipation of treatment affects lagged covariates;
- prior treatment affects the supposedly pretreatment state variable;
- T is short enough for dynamic fixed-effects bias to be important;
- the treatment or outcome process contains heterogeneous dynamics;
- the lag structure is misspecified;
- variables are nonstationary or strongly trending;
- causal relationships are nonlinear or interactive;

- treatment effects vary across units or treatment histories;
- measurement error is substantial;
- there is strong cross-sectional dependence;
- spillovers violate unit-level treatment assumptions;
- researchers are interested in treatment regimes rather than a coefficient on D_t ;
- marginal structural models or g-methods are required to handle treatment-confounder feedback.

The manuscript already recognizes that the recommendation is local to the contemporaneous treatment effect and does not resolve contemporaneous unobserved confounding. The additional limitations should be consolidated into a concise section on **scope conditions**.

7. How Researchers Should Cite or Refer to the Work

Because the manuscript is dated July 2026 and does not identify a journal or working-paper series, the cautious reference is:

Galdino, Manoel, Davi Moreira, and Carolina Dolleães. 2026. "Included Variable Bias in Dynamic Panels: A Diagnostic for Time-Varying Controls and a Benchmark for the Contemporaneous Treatment Effect." Unpublished manuscript.

In prose, researchers should characterize the contribution carefully:

Galdino, Moreira, and Dolleães (2026) propose included variable bias as a linear-regression diagnostic for decomposing the treatment-coefficient shift associated with adding a potentially treatment-responsive control. For contemporaneous treatment effects in TSCS data, they recommend ADL models with fixed effects and substantively justified lagged state variables as a benchmark specification.

Researchers should not cite the paper as establishing that:

- lagging controls always eliminates bad-control bias;
 - ADL plus fixed effects generally identifies causal panel effects;
 - a large IVB value proves that a control should be omitted;
 - the proposed benchmark solves time-varying confounding in general;
 - the IVB decomposition independently identifies whether a variable is a collider, mediator, or confounder.
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8. Editorial and Organizational Recommendations

The manuscript is generally readable, but its two-part contribution—IVB plus the ADL benchmark—is repeated in the abstract, introduction, Sections 2, 4, 5, and conclusion.

Some repetition is useful. In the current version, however, the manuscript sometimes appears to be repeatedly persuading the reader that the diagnostic and benchmark belong together rather than advancing the argument.

A sharper structure would be:

1. motivating application and precise estimand;
2. time-indexed DAGs and formal definitions;
3. IVB as a descriptive decomposition;
4. conditions under which the shift has a causal-bias interpretation;
5. ADL benchmark and identification assumptions;
6. simulations;
7. systematic empirical applications;
8. scope conditions and extensions.

The authors should also replace evaluative labels such as:

- “ADL bad”;
- “ADL safe”;

with neutral labels such as:

- ADL plus contemporaneous Z_t ;
- ADL plus lagged Z_{t-1} ;
- ADL without Z .

“Safe” is too strong because the manuscript itself demonstrates that the specification can remain biased under unobserved confounding and other assumption violations.

9. Suggested Formal Reframing

The paper could benefit from defining three distinct objects.

9.1 Specification Shift

For nested linear projections:

$$\Delta_Z = \beta_{\text{long}} - \beta_{\text{short}} = -\theta_Z \pi_D.$$

This is an algebraic or projection-based quantity. It does not require a causal interpretation.

9.2 Causal Classification

The researcher uses a time-indexed DAG and substantive temporal knowledge to classify Z as:

- a confounder;
- a collider;
- a mediator;
- a precision variable;

- a dual-role variable;
- or a variable whose role is ambiguous.

9.3 Causal Bias

Only after defining a target estimand and imposing identification assumptions should the manuscript write:

$$\text{Bias}(\beta_{\text{long}}) = \beta_{\text{long}} - \beta_{\text{CET}}.$$

This structure would eliminate much of the possible confusion between coefficient movement and causal bias.

It would also strengthen the manuscript's central argument:

IVB does not identify the causal role of a control. It measures the coefficient consequence of conditioning on it, while causal theory determines what that consequence means.

10. Suggested Additional Analyses

The following additions would materially strengthen the paper.

10.1 Sensitivity to Panel Dimensions

Vary N and T to identify when finite- T dynamic panel bias becomes comparable to, or larger than, the bad-control problem being studied.

10.2 Treatment Persistence

Vary the persistence of D_t , not only the persistence of Y_t and Z_t . This matters because lagged treatment may retain associations created by contemporaneous structures.

10.3 Misspecified Lag Length

Compare:

- one-lag ADL;
- two-lag ADL;
- the true lag structure;
- lag structures selected using information criteria.

This would show whether the benchmark is robust to realistic lag misspecification.

10.4 Heterogeneous Dynamics

Allow autoregressive parameters and treatment effects to differ across units. Standard fixed-effects ADL coefficients may be difficult to interpret when dynamics are heterogeneous.

10.5 Common Shocks and Cross-Sectional Dependence

Introduce common factors or correlated errors across units. Many TSCS applications involve global shocks that are not fully captured by time fixed effects.

10.6 Measurement Error

Investigate measurement error in Z_t and Z_{t-1} . Lagging a noisy variable may exacerbate attenuation and alter the treatment-control relationship.

10.7 Nonlinear and Interactive Models

The manuscript briefly discusses nonlinear effects. A stronger assessment would vary:

- nonlinear outcome functions;
- treatment-control interactions;
- heterogeneous treatment effects;
- limited dependent variables.

The authors should clarify whether IVB is primarily a linear-projection diagnostic or whether they envision generalized versions.

10.8 Multiple-Control Simulations

Simulate correlated candidate controls with different causal roles. This would demonstrate how individual and joint IVB diagnostics behave when one variable is a confounder and another is a mediator or collider.

10.9 Comparison with Alternative Causal Methods

The paper could include a limited comparison with:

- marginal structural models;
- g-computation;
- sequential regression;
- structural nested models.

The goal would not be to claim that ADL outperforms these methods generally. It would clarify the difference between a practical linear benchmark and methods designed to address treatment-confounder feedback under stronger identification assumptions.

11. Strengths of the Manuscript

Despite the concerns above, the paper has several important strengths.

11.1 Clear Applied Motivation

The peacekeeping and conflict example effectively demonstrates why controlling for a variable such as conflict intensity may create a dilemma rather than solve one.

11.2 Estimand Discipline

The authors repeatedly limit the positive recommendation to the contemporaneous treatment effect. This is good practice and should be preserved.

11.3 Recognition That the Diagnostic Is Not Automatic

The paper correctly emphasizes that the IVB formula cannot determine whether a control should be included or excluded. This is essential.

11.4 Useful Dual-Role Concept

The idea that the same substantive variable can be a lagged confounder and a contemporaneous collider or mediator is highly relevant to applied research.

11.5 Integration of Theory, Simulation, and Applications

The manuscript does not stop at a formal identity. It uses simulations and published applications to show how the diagnostic might be used.

11.6 Transparent Scope Limitations

The conclusion explicitly acknowledges that the benchmark does not solve contemporaneous unobserved confounding and is not a universal recommendation for all panel estimators.

These strengths provide a solid foundation for revision.

12. Bottom Line

The paper has a strong applied intuition and an effective core message.

Its most compelling contribution is not the equation

$$-\hat{\theta}\hat{\pi}$$

in isolation. It is the integration of:

- time-indexed causal reasoning;
- a transparent decomposition of specification sensitivity;
- a dynamic benchmark tied to a specific estimand.

To reach its potential, the manuscript should:

1. be more conservative and precise about the word **bias**;
2. explicitly state the assumptions required for causal interpretation of the ADL coefficient;
3. address finite- T dynamic panel bias;
4. substantially expand the dynamic DAGs;
5. document the simulation designs completely;
6. develop uncertainty measures for IVB;
7. extend the decomposition to multiple correlated controls;
8. explain the application-selection and causal-coding procedures;
9. make the within-period temporal ordering explicit;
10. distinguish a practical benchmark from a general identification solution.

With these revisions, the manuscript could become a useful political-methodology article and a widely taught guide for auditing time-varying controls in applied panel research.

Recommendation

Major revision

The manuscript has a publishable and potentially influential core, but the conceptual distinction between specification change, causal classification, and causal bias must be formalized before the central claims are fully convincing.